NETWORK SNIFFING

WIRESHARK PACKET ANALYZING

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Network Sniffing using Wireshark [Packet Capture Report]

1. What protocols were seen on the network?

The packet capture contained multiple network protocols including TCP, UDP, ICMP, and ARP. Most traffic was TCP-based communication.

Protocol	Packet Count
UDP	104
ARP	26
ICMP	20
TCP	1220

Fig. Network Protocols

2. Were any credentials or sensitive data visible in plaintext?

Yes. HTTP traffic exposed plaintext information including HTTP headers, session data, and references to default credentials such as admin/password.

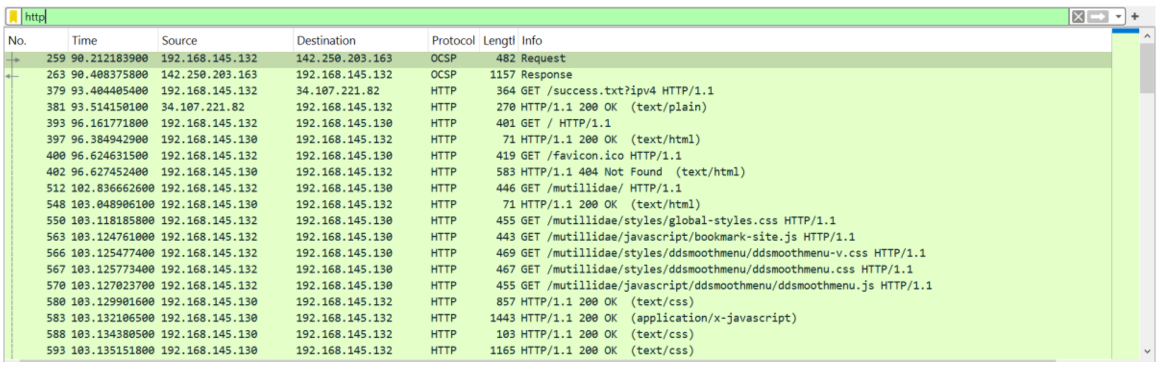

Fig. HTTP Filter

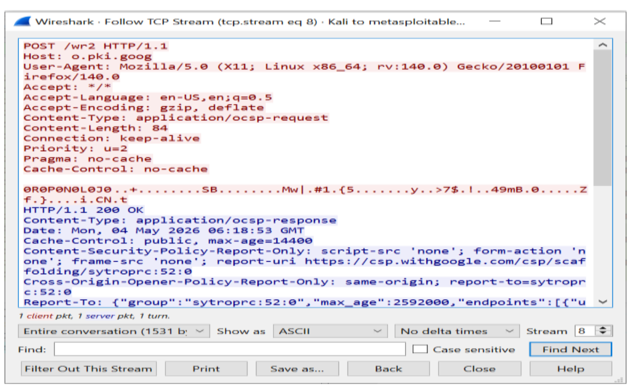

Fig. Plain Text transfer in http

3. What is the likely root cause of the connectivity problem?

ICMP traffic confirmed that hosts could communicate, but some ICMP destination unreachable messages suggest that certain ports or services were unavailable or misconfigured.

4. What solutions do you recommend?

The network should implement encrypted protocols and secure configurations to reduce risk.

- Use HTTPs instead of HTTP
- Swith to SSH instead of Telnet
- Disable unused services and ports
- Use strong authentication
- Apply Firewall and DNS configuration Check

5. Conclusion

The capture demonstrates insecure plaintext communication and possible service misconfiguration. Implementing HTTPS, SSH, firewall hardening, and proper service management would improve security.